



COMPLETE DRAINAGE PROJECT

Survey → Design → Estimate → DPR → Supervision → Construction → Execution



ENGINEERING SCOPE

- 1 Complete drainage planning for cities, tehsils, villages and colonies
- 2 Topographical survey and RL collection for the complete project area
- 3 Hydraulic planning to route runoff safely to the designated main drain / outfall
- 4 Preparation of drawings, estimate, DPR and execution supervision

SURVEY

RL

DESIGN

DPR

SUPERVISION

EXECUTION



AREA, POPULATION & CATCHMENT ASSESSMENT

Project planning begins with understanding the complete drainage area



ENGINEERING SCOPE

- 1 Define project boundary, settlements, roads, colonies and drainage catchments
- 2 Calculate population and development area for the design horizon
- 3 Study natural slopes, low points, ridges, valleys and existing watercourses
- 4 Identify existing drains, outfalls, ponds, culverts and vulnerable waterlogging points
- 5 Prepare catchment-wise drainage planning and proposed flow paths

SURVEY

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COMPLETE TOPOGRAPHICAL SURVEY & RL

Ground levels form the base of the drainage design



ENGINEERING SCOPE

- 1 Establish benchmarks / reference points and record Reduced Levels (RLs)
- 2 Survey roads, plots, drains, culverts, buildings, utilities and natural features
- 3 Take longitudinal levels along proposed drainage alignments
- 4 Collect cross-section levels at required chainages and critical locations
- 5 Prepare contour / spot-level map and direction of natural drainage flow

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LONGITUDINAL SECTION & CROSS-SECTION

Detailed engineering drawings for the proposed drain



ENGINEERING SCOPE

- 1 Prepare complete L-section showing chainage, existing ground RL and proposed invert RL
- 2 Show formation level, bed level, depth, gradient and design water level
- 3 Prepare cross-sections showing top width, bed width, side slopes, depth and lining
- 4 Check widening, depth and hydraulic section against the approved design

L-SECTION / RL CHECK

- 5 Coordinate sections with roads, crossings, culverts and property limits

0 50 100 150
Chainage →

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FLOW DIRECTION, LEVEL STONES & DRAINAGE MARKING

Make the drainage direction and levels clear on site



ENGINEERING SCOPE

- 1 Mark the direction of flow from higher RL to lower RL
- 2 Provide permanent level / benchmark stones at suitable locations
- 3 Mark chainage, drain alignment and important reference levels
- 4 Use visible field markings so site teams can follow the approved alignment

SITE LEVEL MARKING

- 5 Confirm the final outfall / main drain connection and discharge route

RL 102.0

RL 101.2



RL 100.4

SURVEY

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HYDRAULIC CALCULATION & DRAINAGE CAPACITY

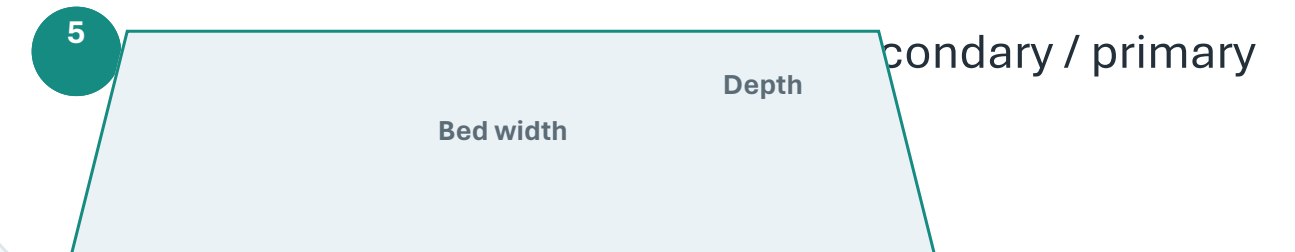
Calculate the runoff and size the drainage network



ENGINEERING SCOPE

- 1 Delineate contributing catchment and estimate design runoff
- 2 Select design flow based on rainfall, catchment characteristics and applicable criteria
- 3 Size drain width, depth, slope and hydraulic section for the design discharge
- 4 Check velocity, freeboard, capacity, crossings and energy conditions

DRAIN SECTION



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ESTIMATE, BOQ & DPR PREPARATION

Convert the survey and design into an executable project



PROJECT DOCUMENTATION

ENGINEERING SCOPE

- 1 Prepare quantity calculations from L-sections and cross-sections
- 2 Prepare item-wise estimate and Bill of Quantities (BOQ)
- 3 Include excavation, disposal, lining/RCC/masonry, culverts and structures as required
- 4 Prepare drawings, specifications, methodology, cost and implementation schedule
- 5 Compile the complete Detailed Project Report (DPR) for approval and execution

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CONSTRUCTION SUPERVISION & QUALITY CONTROL

Ensure the drain is built exactly as designed



QUALITY CONTROL

ENGINEERING SCOPE

- 1 Set out alignment, chainage and RLs before construction
- 2 Check excavation width, depth, formation level and drain gradient
- 3 Verify cross-section, widening, lining thickness and structural dimensions
- 4 Supervise RCC / masonry, bedding, joints, curing and finishing
- 5 Record measurements, inspections, tests, deviations and corrective actions

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EXECUTION, TESTING & FINAL DRAINAGE CONNECTION

From construction to commissioning



ENGINEERING SCOPE

- 1 Execute drain sections, crossings, outfalls and protection works in sequence
- 2 Check continuity of flow and remove construction obstructions
- 3 Verify invert levels and final connection to the main drain / natural outfall
- 4 Prepare as-built levels, drawings and completion records
- 5 Hand over the functional drainage system with maintenance requirements

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INTEGRATED PROJECT DELIVERY

A complete end-to-end drainage service



ENGINEERING SCOPE

- 1 Survey the complete area and establish reliable RL / benchmark control
- 2 Prepare catchment plan, L-section, cross-sections and drainage drawings
- 3 Complete hydraulic calculations, estimate, BOQ and DPR
- 4 Supervise construction and verify dimensions, levels and workmanship



SURVEY

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